CNN (convolutional Neural Network)

visual data process, classify

Human brain

- visual cortex

V_1 - orientation, edges, lines

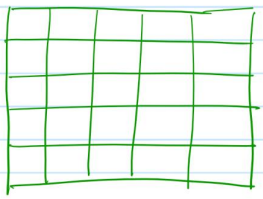
V_2 - texture and complex shapes

V_3 - motion and dynamic form of the shapes

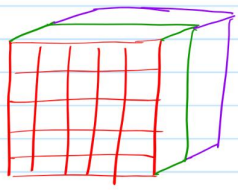
V_4 - color and form, object recognition

V_5 - motion perception, moving object tracking, speed & direction

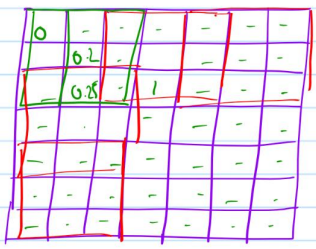
Gray scale
(White/Black)
Black = 0,
White = 255



RGB Image
Red
Green
Blue



Convolutional operation



6 x 6

Filter = 3x3

+1	0	-1
+2	0	-2
+1	0	-1

- Horizontal

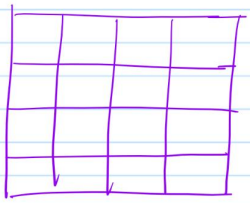
+1	+2	+1
0	0	0
-1	-2	-1

- vertical

3x3

stride - 2

o/p



4x4

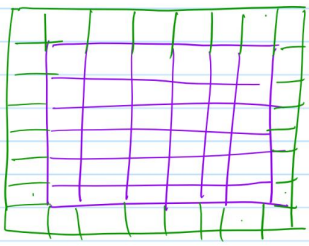
$o/p = (n-f)+1$

$(6-3)+1 \Rightarrow 4$

28×28 $\quad$ 3×3 $\quad$ $(28-3)+1 \Rightarrow 26$

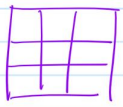
6×6 $\quad$ $(28-6)+1 \Rightarrow 23$

Padding Technique

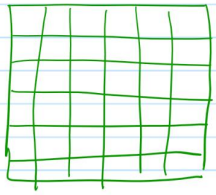


6×6

8×8



3×3



6×6

$n - f + 1$

$8 - 3 + 1 = 6$

To calculate padding size

$n - f + 2p + 1 = 6$

$6 - 3 + 2p + 1 = 6$

$3 + 2p + 1 = 6$

$4 + 2p = 6$

$2p = 6 - 4$

$p = 2/2 = 1$

Convolutional layer

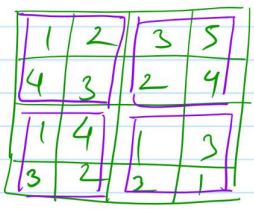
Conv. + Relu $\longrightarrow$

$\boxed{6 \times 6}$ $\quad$ $\boxed{3 \times 3}$ $\quad$ $\boxed{4 \times 4}$

single conv. layer.

pulling Technique.

① max pulling ② min pulling ③ mean/Avg. pulling



max

4	5
4	3

min

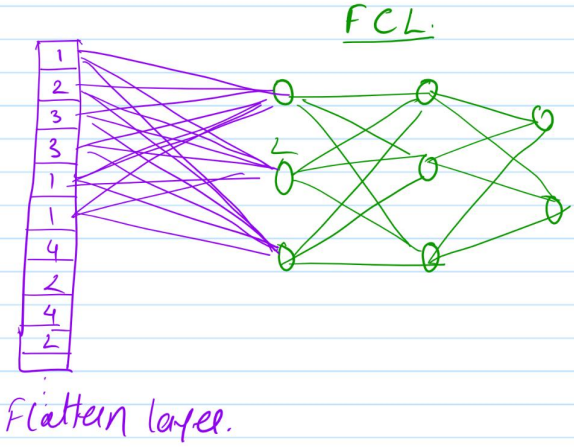
1	2
1	1

2.5	3.5
2.5	1.75

Avg.

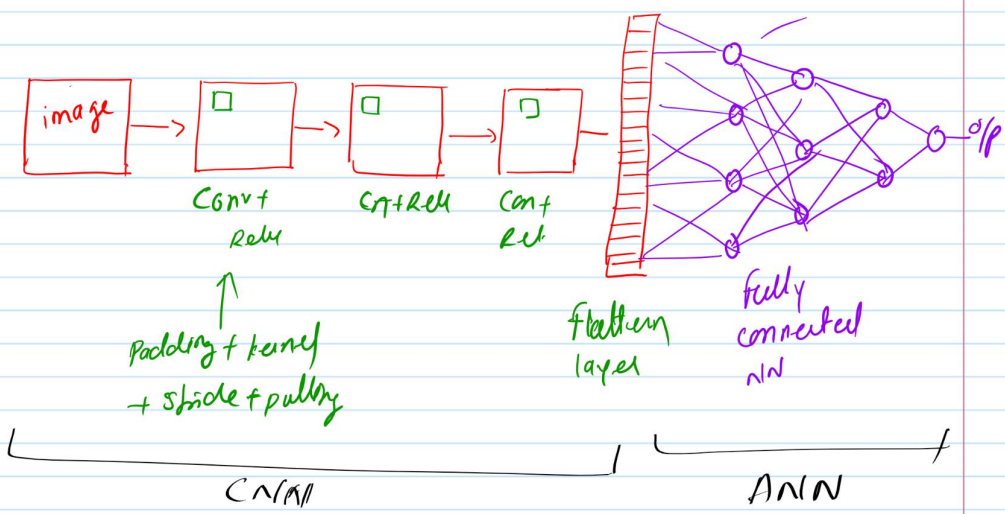
flatten layer

1	2	3	3
1	1	4	2
1	4	2	4
2	4	3	1



CNN

1) conv. layer
2) kernel / filter
3) stride
4) Padding
5) pulling
6) flatten
7) Fully conn. layer
8) Dropout
9) Batch normalization.



famous CNN architecture

1. Lenet - 5 - 1998, 32×32, Grey scale imag.
2. Alexnet
3. VGG - 16, 19
4. Inception / GoogleNet
5. ReNet (microsoft)